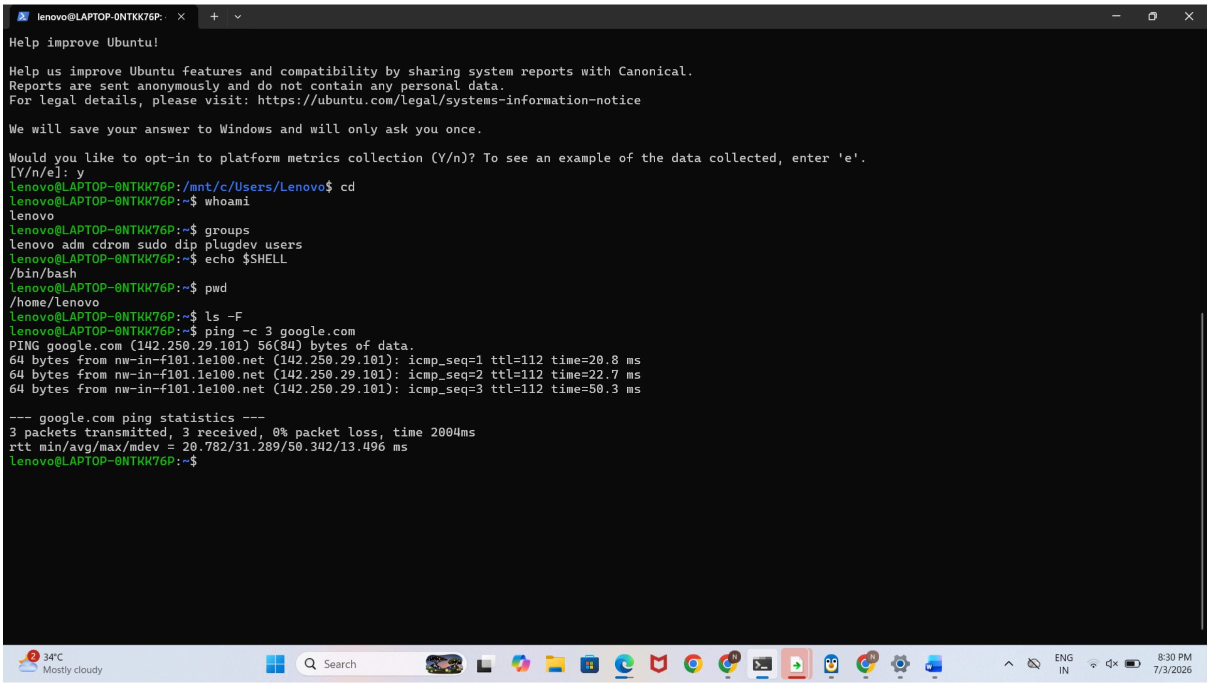
Question 1 Summary: Linux Environment Verification

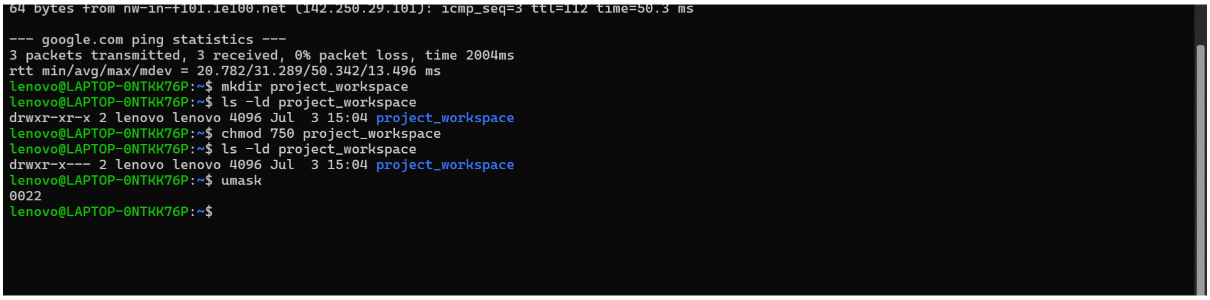
To verify my Linux environment, I executed a series of diagnostic commands. I used `whoami` to confirm my user identity as `lenovo`, and `groups` to list my security permissions, which confirmed my administrative group membership. I verified my shell environment using `echo $SHELL`, which returned `/bin/bash`. I confirmed my file system location with `pwd` and listed



files using `ls -F` to ensure a clean workspace. Finally, I utilized `ping -c 3 google.com` to verify that my WSL environment has active network connectivity and that the system is properly configured for internet communication.

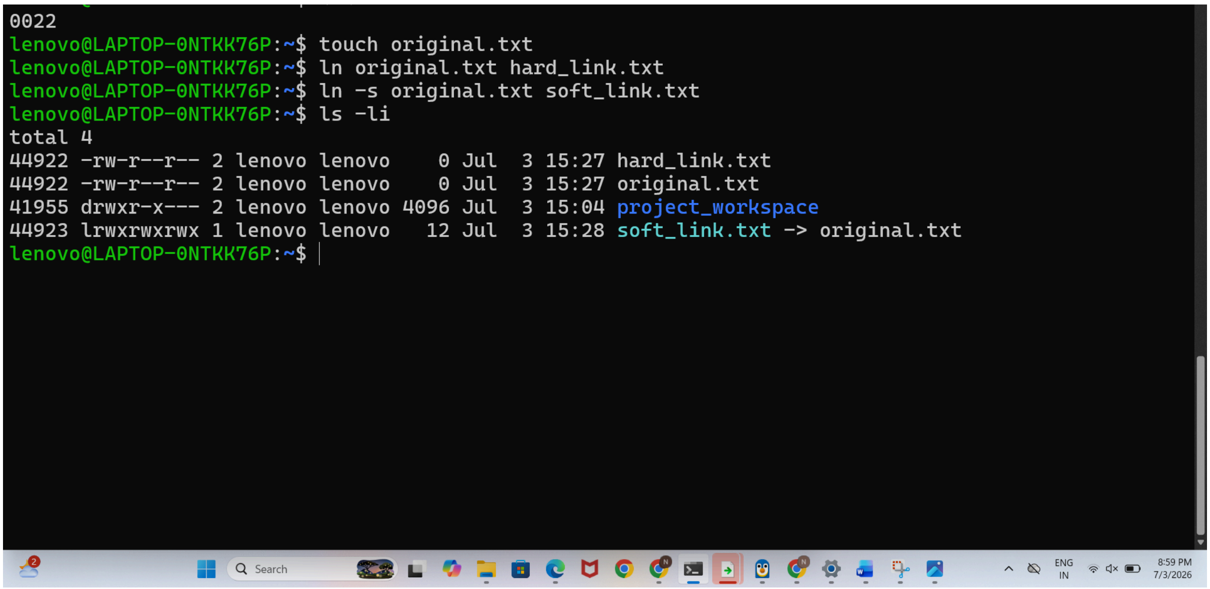
Question 2 Summary: Secure Project Workspace Setup

To establish a secure working environment, I created a new directory named `project_workspace`. I inspected its initial permissions using `ls -ld` to establish a baseline. I then applied the `chmod 750` command, which restricts access so that only the owner has full read, write, and execute permissions, while group members have read and execute access, and "others" have zero access. This configuration ensures that sensitive project data is shielded from unauthorized users. Finally, I reviewed the system `umask` to confirm the default permission mask that will govern the security of all future files and directories I create within this session.



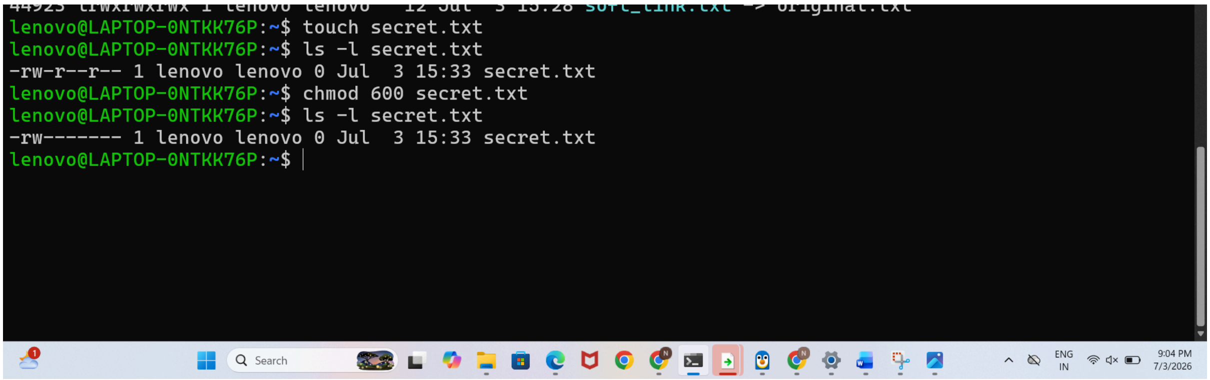
Question 3 Summary: Hard and Symbolic Links

I created an `original.txt` file and established both a hard link and a symbolic (soft) link. Using the `ls -li` command, I verified that `hard_link.txt` shares the same inode number as the original, meaning it is essentially a second name for the same data. In contrast, `soft_link.txt` has a unique inode and acts as a pointer to the original filename, demonstrating how symbolic links function as shortcuts.



Question 4 Summary: File Privacy

I created `secret.txt` and checked its initial permissions using `ls -l`. I then applied `chmod 600` to the file, which modifies the permission bits so that only the owner has read and write access, while strictly removing all access for group members and others. This configuration is a critical security practice for ensuring data privacy and protecting sensitive files from unauthorized access.



Question 5 Summary: Background Process Management

I initiated a background process using `sleep 100 &`, which allows the command to continue running without blocking the terminal. I verified its active status using `ps aux | grep sleep`, which displays the process ID (PID). After confirmation, I used `kill %1` to terminate the job by its job ID. A final check with `ps aux | grep sleep` confirmed that the process was successfully removed from the system, demonstrating effective background process control.